

Which Public Transport Modes Are Most Used in Victoria and Where Should Investment Be Focused?

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1 Executive Summary

2 Introduction

3 Methodology

```
monthly_raw <- read_csv("data/raw/74174b02-76bc-4d10-ae7c-401d90ef033c.csv")
stations_raw <- read_csv("data/raw/_published_public_transport_annual_metropolitan_train_sta
```

```

monthly_clean <- monthly_raw |>
  clean_names() |>
  mutate(across(c(metropolitan_train, metropolitan_tram, metropolitan_bus,
                  regional_train, regional_coach, regional_bus),
    ~ parse_number(as.character(.)))) |>
  pivot_longer(
    cols = c(metropolitan_train, metropolitan_tram, metropolitan_bus,
             regional_train, regional_coach, regional_bus),
    names_to = "mode",
    values_to = "boardings"
  )

```

```

stations_clean <- stations_raw |>
  clean_names() |>
  filter(fin_year == "FY24-25") |>
  mutate(across(c(pax_annual, pax_weekday, pax_norm_weekday, pax_sch_hol_weekday,
                  pax_saturday, pax_sunday, pax_pre_am_peak, pax_am_peak,
                  pax_interpeak, pax_pm_peak, pax_pm_late),
    ~ parse_number(as.character(.))))

```

```

write_csv(monthly_clean, "data/processed/monthly_clean.csv")
write_csv(stations_clean, "data/processed/stations_clean.csv")

```

```

mode_summary <- monthly_clean |>
  group_by(mode) |>
  summarise(total_boardings = sum(boardings, na.rm = TRUE), .groups = "drop") |>
  arrange(desc(total_boardings))
mode_summary

```

```

# A tibble: 6 x 2
  mode                total_boardings
  <chr>                <dbl>
1 metropolitan_train  1366415976
2 metropolitan_tram   1128683201
3 metropolitan_bus     797593599
4 regional_train      146151590
5 regional_bus         92905490
6 regional_coach       8709111

```

```
top_stations <- stations_clean |>
  arrange(desc(pax_annual)) |>
  select(stop_name, pax_annual, pax_weekday, pax_saturday, pax_sunday) |>
  slice_head(n = 10)
top_stations
```

A tibble: 10 x 5

	stop_name <chr>	pax_annual <dbl>	pax_weekday <dbl>	pax_saturday <dbl>	pax_sunday <dbl>
1	Flinders Street	19633300	60900	43150	31350
2	Southern Cross	14696050	47600	26500	19550
3	Melbourne Central	11876900	38300	20650	17250
4	Parliament	6344650	22350	6450	4750
5	Footscray	4383950	14000	8850	5800
6	Richmond	3712750	9850	13000	8850
7	Flagstaff	3445200	12350	2900	2200
8	Caulfield	3205100	9600	8250	5600
9	South Yarra	3199050	10350	5900	4100
10	Box Hill	2648350	8350	5350	4100

```
ggplot(mode_summary, aes(x = reorder(mode, total_boardings), y = total_boardings, fill = mode)) +
  geom_col(show.legend = FALSE) +
  coord_flip() +
  scale_y_continuous(labels = comma) +
  labs(x = NULL, y = "Total boardings")
```

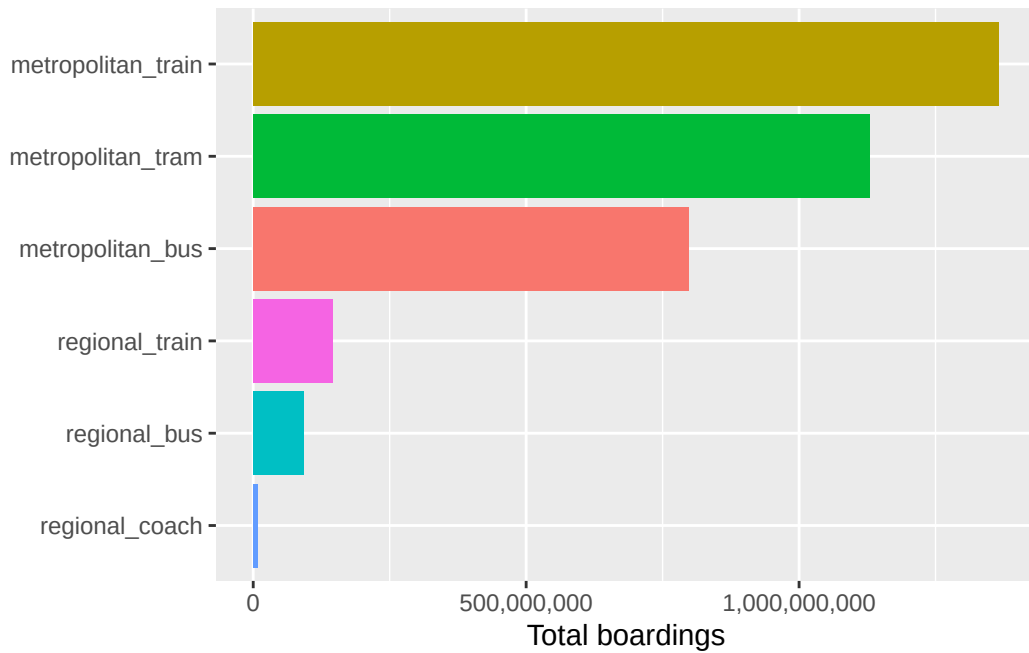


Figure 1: Total public transport boardings by mode across the monthly Victorian dataset.

4 Results

5 Discussion, Conclusion and Recommendations

6 References

1. Monthly public transport patronage by mode. Victorian open data. [link](#).
2. Annual metropolitan train station patronage (station entries). Victorian open data. [link](#).